

Integer Spectra Suffice for Maximal Separable-Sum Multiplicities

Automated Math Discovery Candidate Proof

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Status. Candidate full solution, likely valid pending expert review. The proof gives a pointwise strengthening of Conjecture 1.4: every real matrix can be replaced by an integer matrix preserving the complete multiplicity pattern through any prescribed finite spectral level.

Source and Conjecture

This packet concerns Conjecture 1.4 on page 4 of Helffer, Hoffmann-Ostenhof, and Marquetand, *On maximal multiplicities for Hamiltonians with separable variables* [1], arXiv:1908.02752.

For a positive integer N , let $\mathfrak{M}(N)$ consist of N rows

$$a_1^n < a_2^n < a_3^n < \cdots, \quad 1 \leq n \leq N,$$

under the source paper's standing assumption that the associated sum spectrum is discrete and tends to $+\infty$. The spectrum is formed from all tuple sums

$$S_{\mathfrak{A}}(j_1, \dots, j_N) = \sum_{n=1}^N a_{j_n}^n.$$

Ordering with multiplicity gives eigenvalues $\lambda_k(\mathfrak{A})$, and $m(k, \mathfrak{A})$ is the full number of representations of $\lambda_k(\mathfrak{A})$. The maximal possible value is

$$\mathfrak{m}_k(N) = \sup_{\mathfrak{A} \in \mathfrak{M}(N)} m(k, \mathfrak{A}).$$

Conjecture 1.4 states:

For determining $\mathfrak{m}_k(N)$, it suffices to take the supremum over the $\mathfrak{A}$ with all $a_i^j \in \mathbb{Z}$.

Theorem 1.3 *For any $k \in \mathbb{N}^*$ and $N \in \mathbb{N}^*$, we have*

$$\mathbf{m}_k(N) \leq \mathbf{m}_{k+1}(N). \quad (1.7)$$

Note that we cannot hope to have always a strict inequality in (1.7) (see below the results for $N = 3$).

As an intermediate result, it could be useful to validate the following conjecture:

Conjecture 1.4 *For determining $\mathbf{m}_k(N)$, it suffices to take the supremum in (1.6) over the $\mathfrak{A}$ with the $a_i^j \in \mathbb{Z}$.*

This will indeed permit to use a computer for determining $\mathbf{m}_k(N)$ by looking at a finite number of possibilities. But the size of the computations could increase dramatically with k and N . Actually, all the numerical results we will be referring to in this paper are done by computation involving only integers.

We do not give in this introduction a general conjecture. Our initial guess was that the matrix associated with the radial harmonic oscillator was a maximizer for any k . Although this remark is quite useful for getting a first

Proof Intuition

Although each row is infinite, a fixed spectral level only sees finitely many entries. After translating every row so that its first entry is zero, all entries are nonnegative. If $\Lambda = \lambda_k(\mathfrak{A})$, then any row entry exceeding Λ cannot occur in a representation at or below Λ .

On the resulting finite box of index tuples, the spectral pattern is specified by finitely many integer-linear equalities and strict integer-linear inequalities between tuple sums. The equalities cut out a rational linear subspace. The inequalities select a relatively open chamber inside it. Rational points are dense in a rational subspace, so the chamber contains a rational point with exactly the same equalities and strict order relations. Clearing denominators gives integer row prefixes. Appending integer tails above the preserved target level prevents any new tuple from entering the finite spectral pattern.

Main Result

Theorem 1 (Pointwise integer realization). *Let $N, k \geq 1$, and let $\mathfrak{A} \in \mathfrak{M}(N)$ have real entries. There exists $\mathfrak{B} \in \mathfrak{M}(N)$ all of whose entries are integers such that*

$$m(k, \mathfrak{B}) = m(k, \mathfrak{A}).$$

More strongly, the equality classes and strict ordering of the spectral multiset of $\mathfrak{A}$ up to and including the entire equality class containing $\lambda_k(\mathfrak{A})$ are preserved by $\mathfrak{B}$.

Consequently, for every $N, k \geq 1$,

$$\sup_{\mathfrak{A} \in \mathfrak{M}(N)} m(k, \mathfrak{A}) = \sup_{\substack{\mathfrak{A} \in \mathfrak{M}(N) \\ a_j^n \in \mathbb{Z} \ \forall j, n}} m(k, \mathfrak{A}),$$

so Conjecture 1.4 has an affirmative answer.

The Rational-Chamber Lemma

Lemma 1. *Let $x \in \mathbb{R}^d$, and let $\ell_1, \dots, \ell_s$ and $h_1, \dots, h_t$ be linear forms on $\mathbb{R}^d$ with integer coefficients. Assume*

$$\ell_i(x) = 0 \quad (1 \leq i \leq s), \quad h_j(x) > 0 \quad (1 \leq j \leq t).$$

Then there is $q \in \mathbb{Q}^d$ satisfying the same conditions. Moreover, q may be chosen arbitrarily close to x within $V = \bigcap_i \ker \ell_i$.

Proof. The subspace V is the kernel of a matrix with integer entries. Gaussian elimination over $\mathbb{Q}$ gives a basis $v_1, \dots, v_r$ of V with rational coordinates. Write $x = \sum_i c_i v_i$. Approximating the real coefficients c_i by rationals produces rational points dense in V . The conditions $h_j > 0$ define a relatively open neighborhood of x in V , so one of these rational points lies in that neighborhood. $\square$

Proof of Theorem 1

Subtract a_1^n from every entry in row n . This translates the complete spectrum by the constant $-\sum_n a_1^n$ and changes no representation multiplicity or spectral ordering. We may therefore assume

$$a_1^n = 0, \quad 0 < a_2^n < a_3^n < \dots \quad (1 \leq n \leq N). \quad (1)$$

Put $\Lambda = \lambda_k(\mathfrak{A})$. For each row define

$$r_n = \max\{j : a_j^n \leq \Lambda\}.$$

These integers are finite by the standing discreteness assumption, and are at least one because the first entries are zero. Set

$$I = \prod_{n=1}^N \{1, \dots, r_n\}.$$

Every tuple whose sum is at most Λ belongs to I : if $j_n > r_n$ for some n , then $a_{j_n}^n > \Lambda$, while all other summands are nonnegative by (1).

Now regard the finite collection of prefix entries

$$a = (a_j^n)_{1 \leq n \leq N, 1 \leq j \leq r_n}$$

as a point of $\mathbb{R}^d$, where $d = \sum_n r_n$. For $u, v \in I$, define the integer-coefficient linear form

$$L_{uv}(z) = \sum_{n=1}^N z_{u_n}^n - \sum_{n=1}^N z_{v_n}^n.$$

Let $V \subset \mathbb{R}^d$ be cut out by the equations

$$z_1^n = 0 \quad (1 \leq n \leq N), \quad L_{uv}(z) = 0 \quad \text{whenever } L_{uv}(a) = 0. \quad (2)$$

This is a rational linear subspace containing a .

Inside V , impose the finitely many strict conditions

$$z_{j+1}^n - z_j^n > 0 \quad (1 \leq n \leq N, 1 \leq j < r_n), \quad (3)$$

and

$$\text{sgn}(L_{uv}(a))L_{uv}(z) > 0 \quad (u, v \in I, L_{uv}(a) \neq 0). \quad (4)$$

Here $\text{sgn}(L_{uv}(a))$ is 1 or -1 , so every displayed condition uses an integer-coefficient linear form. Conditions (3)–(4) define a relatively open chamber in V which contains a . Lemma 1 supplies a rational point q in this chamber.

Choose a positive common denominator D of all coordinates of q and put

$$b_j^n = Dq_j^n \quad (1 \leq j \leq r_n).$$

The b_j^n are integers. Equations (2) preserve every equality between tuple sums indexed by I , while (4) preserves the strict order of every unequal pair. Thus the two finite multisets

$$\{S_{\mathfrak{A}}(u) : u \in I\}, \quad \{S_{\mathfrak{B}}(u) : u \in I\}$$

have identical ordered equality-class patterns. In particular, if $u^* \in I$ represents Λ , then

$$T := S_{\mathfrak{B}}(u^*)$$

is independent of the chosen representation u^* of Λ , and its equality class has the same cardinality as that of Λ .

It remains only to complete the rows without disturbing this finite pattern. For $s \geq 1$, define

$$b_{r_n+s}^n = \max\{T, b_{r_n}^n\} + s, \quad 1 \leq n \leq N. \quad (5)$$

Every row is now a strictly increasing, unbounded integer sequence beginning at zero. Any tuple outside I uses at least one tail entry, which by (5) is strictly larger than T ; all its other entries are nonnegative. Hence every such tuple sum is larger than T .

It follows that the complete multiplicity-counted ordering through the class of T for $\mathfrak{B}$ is exactly the pattern already preserved on I . The k -th term lies in the corresponding equality class, and therefore

$$m(k, \mathfrak{B}) = m(k, \mathfrak{A}).$$

This proves the pointwise assertion. Since integer matrices form a subclass of real matrices, one inequality between the two suprema is immediate; the pointwise construction gives the reverse inequality. The theorem follows.

Verification Notes and Scope

- The proof preserves the ordering *with multiplicity*; it is unaffected by the source paper's occasional distinct-value phrasing.
- No compactness or attainment of the supremum is used. The comparison is pointwise for every admissible real matrix.
- The appended tails are essential: rationalizing only the visible prefixes would not by itself define an element of $\mathfrak{M}(N)$.

- The theorem uses precisely the source paper’s standing discrete, unbounded spectral model. The claim does not address arbitrary bounded strictly increasing sequences, for which the source’s enumeration tending to $+\infty$ would fail.
- No computer-assisted step is part of the proof.

Bounded Novelty Check

On July 22, 2026, the run’s registry, solution, attempt, proof-gap, title, and source-ranked indexes were searched for arXiv:1908.02752, the exact paper title, and combinations of “integer entries,” “maximal multiplicities,” “separable variables,” and “rational chamber.” No prior answer was found. Bounded web searches used the exact title, arXiv id, author names, the wording of Conjecture 1.4, and close variants of “maximal multiplicities integer spectrum.” They found the source paper, an author publication list, and unrelated spectral-multiplicity papers, but no direct answer or later paper resolving Conjecture 1.4. Novelty confidence is moderate pending specialist review.

Human Review Recommendation

This packet is recommended for expert review as a candidate full solution. The main review points are the finiteness of each visible prefix under the source’s admissibility convention, the fact that all finite equality data define a rational subspace, and the exclusion of every appended-tail tuple from the preserved spectral range.

References

- [1] B. Helffer, T. Hoffmann-Ostenhof, and P. Marquetand, *On maximal multiplicities for Hamiltonians with separable variables*, arXiv:1908.02752 (2019).